CYBERSECURITY TESTING LAB SETUP USING VIRTUALBOX

Project: WK1-PM1 – Lab Setup: VirtualBox and Kali Linux
Subject: Cybersecurity and Ethical Hacking Practice
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Date: 31-10-2026

1. Introduction

The purpose of this practical assignment was to set up a cybersecurity testing lab environment on a personal computer using Oracle VirtualBox. The lab provides an isolated environment where different operating systems can communicate and where cybersecurity and ethical hacking exercises can be performed safely.

The main objective was to install and configure Kali Linux as the attacking machine, establish a custom NAT Network using the 10.0.0.0/24 subnet, configure internet connectivity, enable communication features between the host and virtual machines, and create a shared folder.

As an additional exercise, Windows 10 was installed as a second virtual machine to act as a target machine for network testing.

2. Tools and Technologies Used

The following tools and technologies were used during the setup:

Tool / Technology	Purpose
Linux Mint	Main host operating system
Oracle VirtualBox	Virtualization platform
Kali Linux	Attacking and cybersecurity testing machine
Windows 10	Target and testing machine
NAT Network	Network communication between virtual machines
NetworkManager (nmcli)	Kali network configuration

Command Prompt	Windows network configuration and testing
Terminal	Executing Linux commands

3. Lab Network Architecture

The lab was configured using a custom VirtualBox NAT Network named NatNetwork.

The network uses the following configuration:

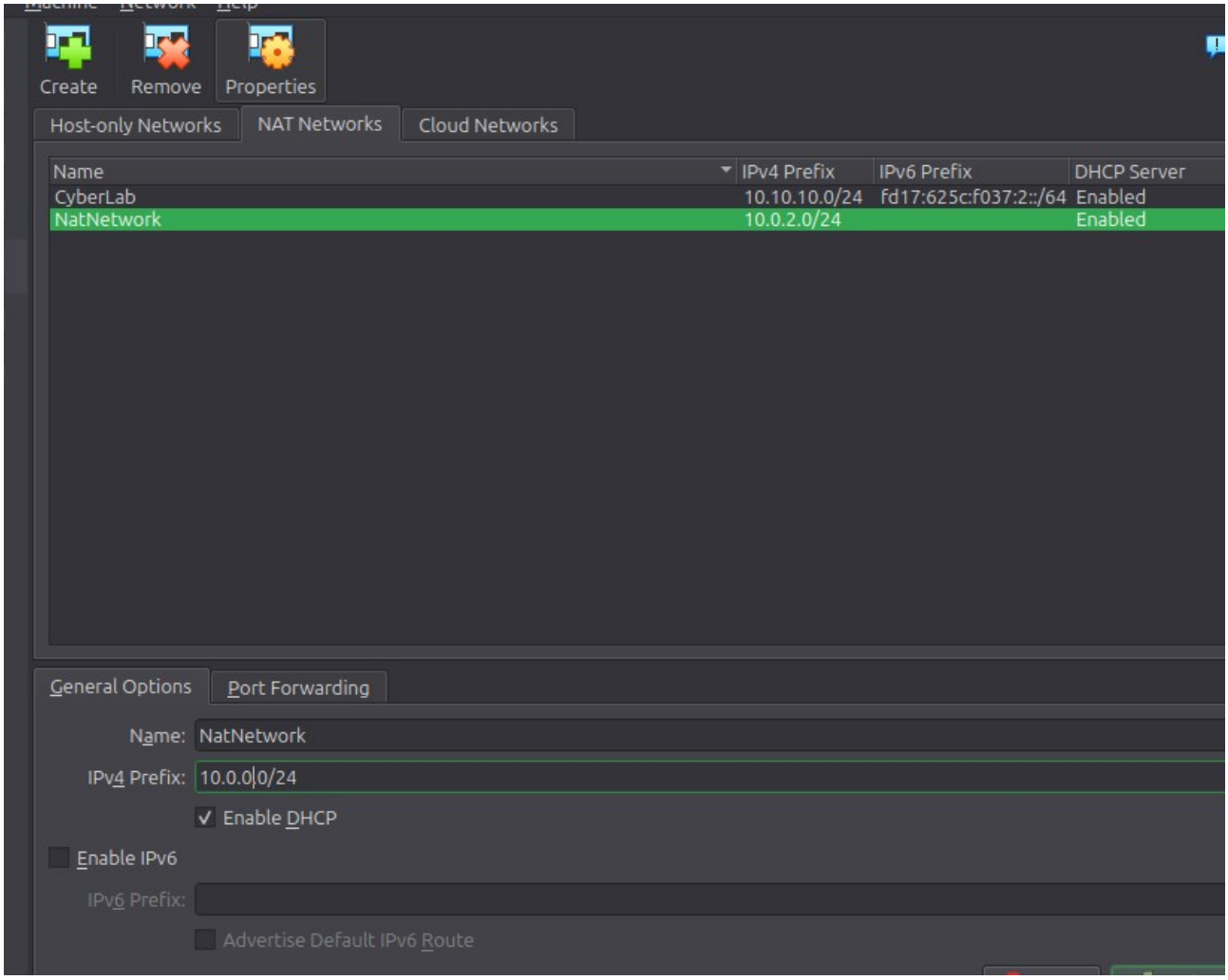
Network setting	Configuration
Network name	NatNetwork
Network type	NAT Network
IPv4 network	10.0.0.0/24
Gateway	10.0.0.1
DHCP	Enabled
IPv6	Disabled

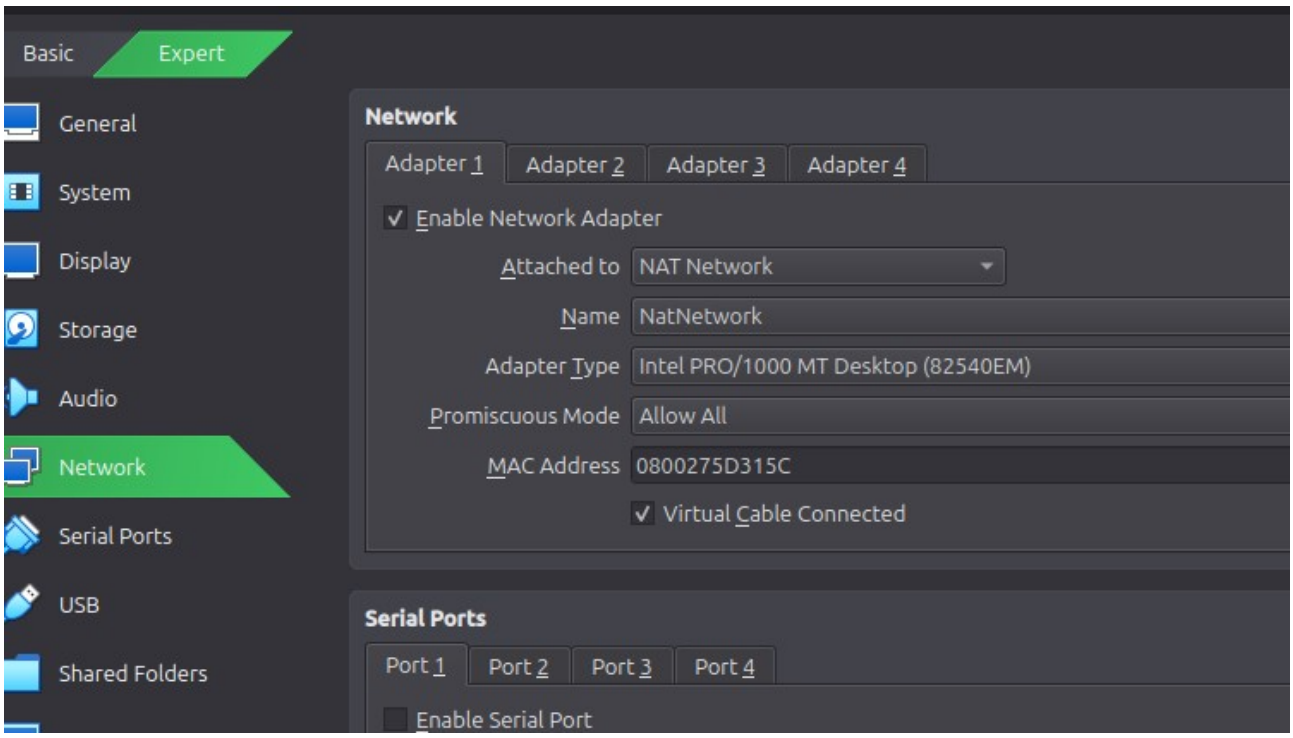
The two virtual machines were connected to the same NAT Network, allowing them to communicate with one another.

IP addressing plan

Device	Role	IP address
Linux Mint	Host operating system	Existing host network address
Kali Linux	Attacking machine	10.0.0.2/24
Windows 10	Target machine	10.0.0.10/24
NAT Network gateway	Network gateway	10.0.0.1

The other virtual machines shown in the original assignment diagram, such as Windows 11, Windows 7, Windows Server 2016, and Android, were optional. They were not installed as part of this setup.



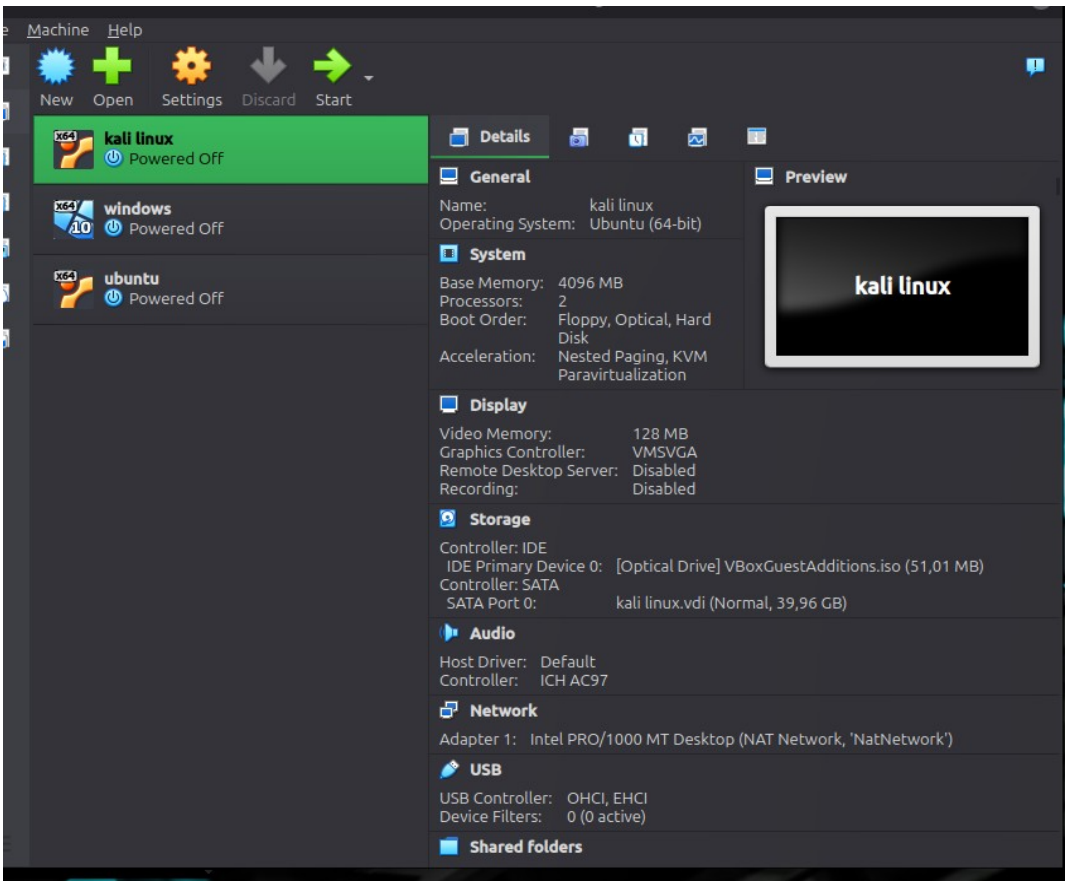


4. Phase One: Kali Linux Installation and Configuration

4.1 VirtualBox installation

Oracle VirtualBox was used as the base virtualization platform. It allows multiple operating systems to run as virtual machines on the same physical computer without requiring separate physical machines.

Kali Linux was installed inside VirtualBox rather than directly on the computer's hardware. This allows the Kali environment to operate independently while using the host computer's resources.

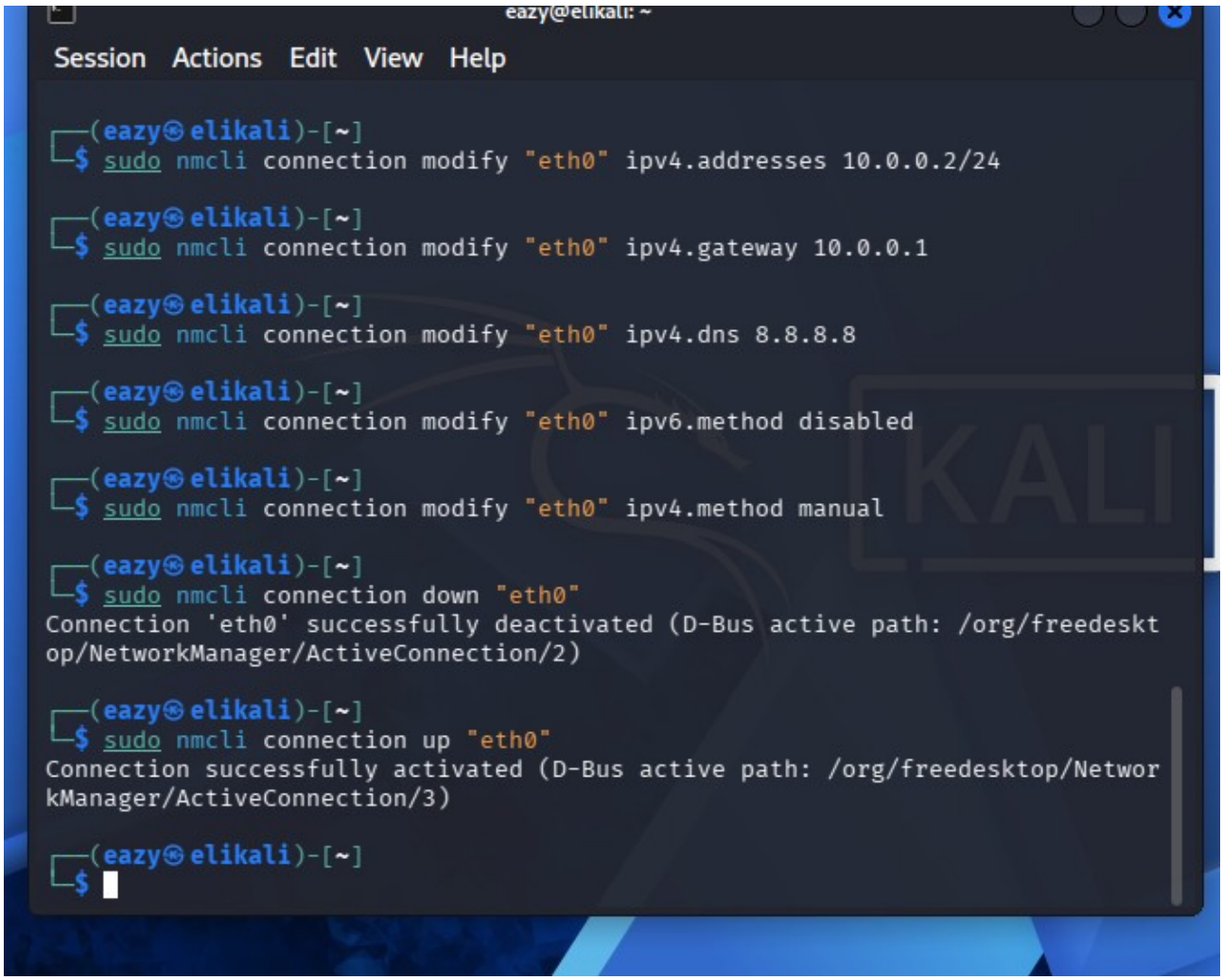


4.2 Creating the NAT Network

A custom NAT Network was created in VirtualBox to allow the virtual machines to communicate with each other while retaining network access through the host.

The network was named NatNetwork, and its IPv4 prefix was configured as 10.0.0.0/24. DHCP was enabled, while IPv6 was disabled.

This configuration established the network environment required for the lab.



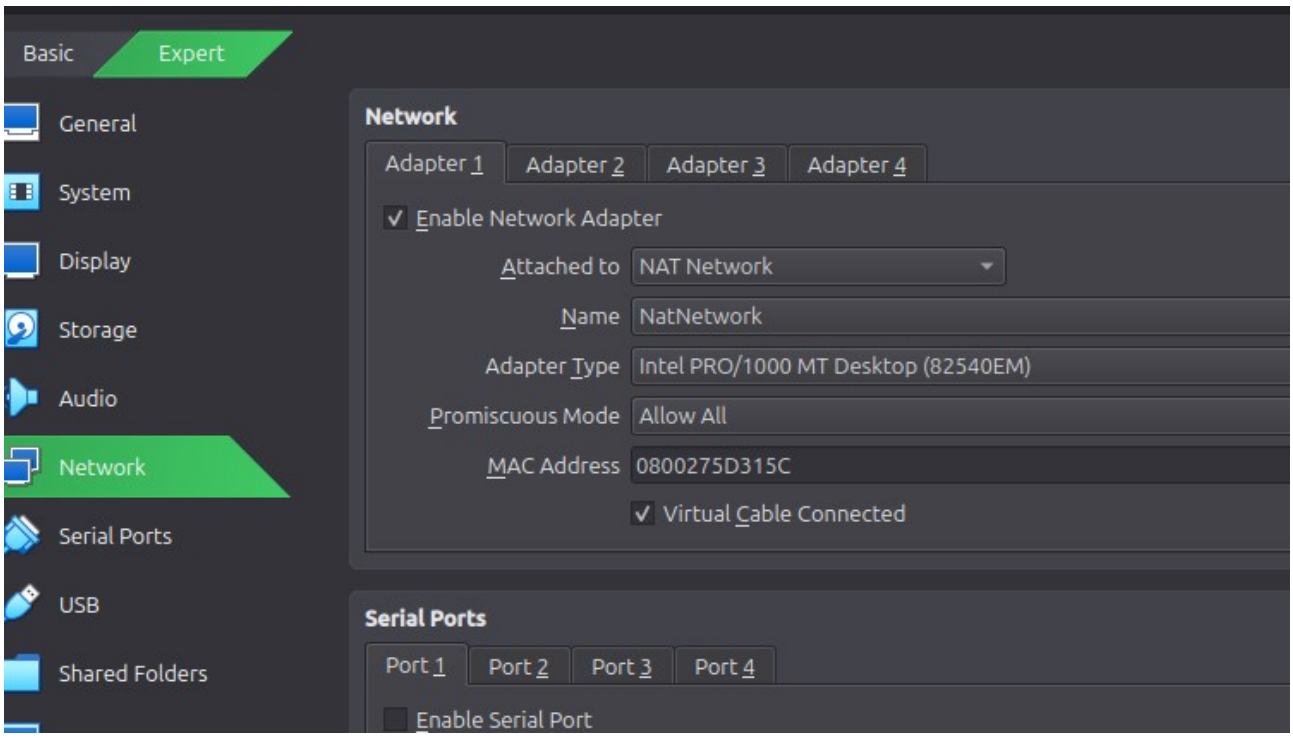
4.3 Configuring the Kali Linux network adapter

The Kali Linux virtual machine was configured to use the custom NAT Network.

The network adapter settings were configured as follows:

- Adapter 1 was enabled.
- Attached to: NAT Network.
- Network name: NatNetwork.
- Promiscuous Mode: Allow All.
- Cable Connected: Enabled.

This allowed Kali Linux to participate in the custom lab network.



4.4 Configuring the static IP address

Kali Linux was assigned the static IP address 10.0.0.2/24, as required by the assignment.

The network configuration was:

Setting	Value
Method	Manual
IP address	10.0.0.2
Netmask	255.255.255.0
Gateway	10.0.0.1
DNS server	8.8.8.8

The configuration was applied using NetworkManager through the Kali terminal.

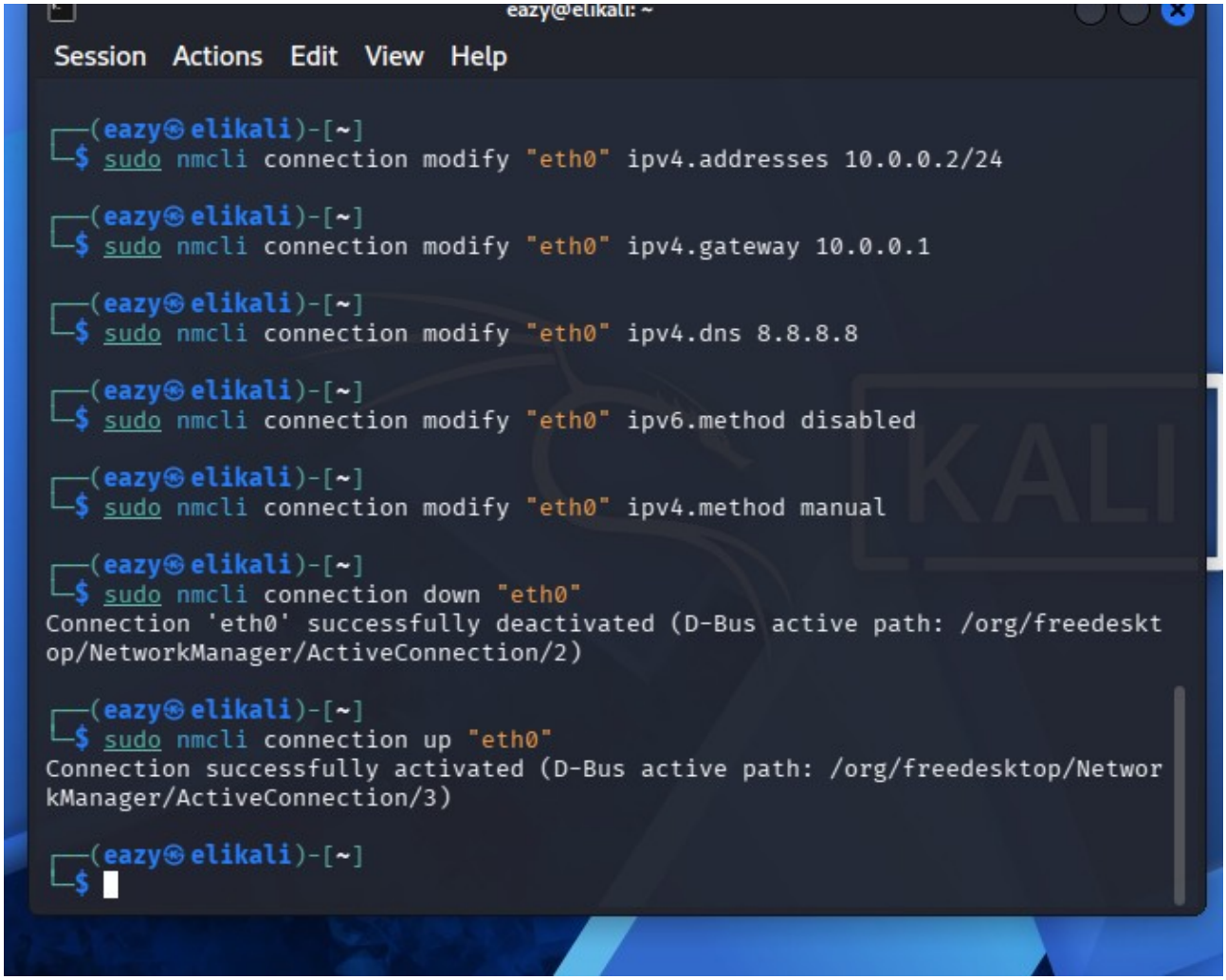
The commands used were:

```
sudo nmcli connection modify "eth0" \
ipv4.method manual \
ipv4.addresses 10.0.0.2/24 \
ipv4.gateway 10.0.0.1 \
ipv4.dns 8.8.8.8 \
ipv6.method disabled
```

The network connection was then restarted:

```
sudo nmcli connection down "eth0"

sudo nmcli connection up "eth0"
```



4.5 Resolving internet connectivity issues

During the setup, Kali Linux initially experienced internet connectivity problems.

The network configuration was adjusted until internet access was restored. The assignment document also provided troubleshooting commands for VirtualBox 7 and newer Kali versions:

```
sudo nmcli connection modify "eth0" ipv4.dad-timeout 0
```

```
sudo nmcli connection down "eth0"
```

```
sudo nmcli connection up "eth0"
```

4.6 Configuring shared folders

A shared folder was configured to allow files to be exchanged between the Linux Mint host and Kali Linux.

The host directory used was:

```
/home/eli/Downloads
```

The folder was added through VirtualBox's Shared Folders settings with the following options:

- Folder name: Downloads
- Access: Read and Write
- Auto-mount: Enabled
- Make Permanent: Enabled

Inside Kali Linux, the shared folder became accessible through:

```
/media/sf_Downloads
```

The folder was tested by creating a file inside Kali Linux:

```
touch /media/sf_Downloads/kali_test.txt
```

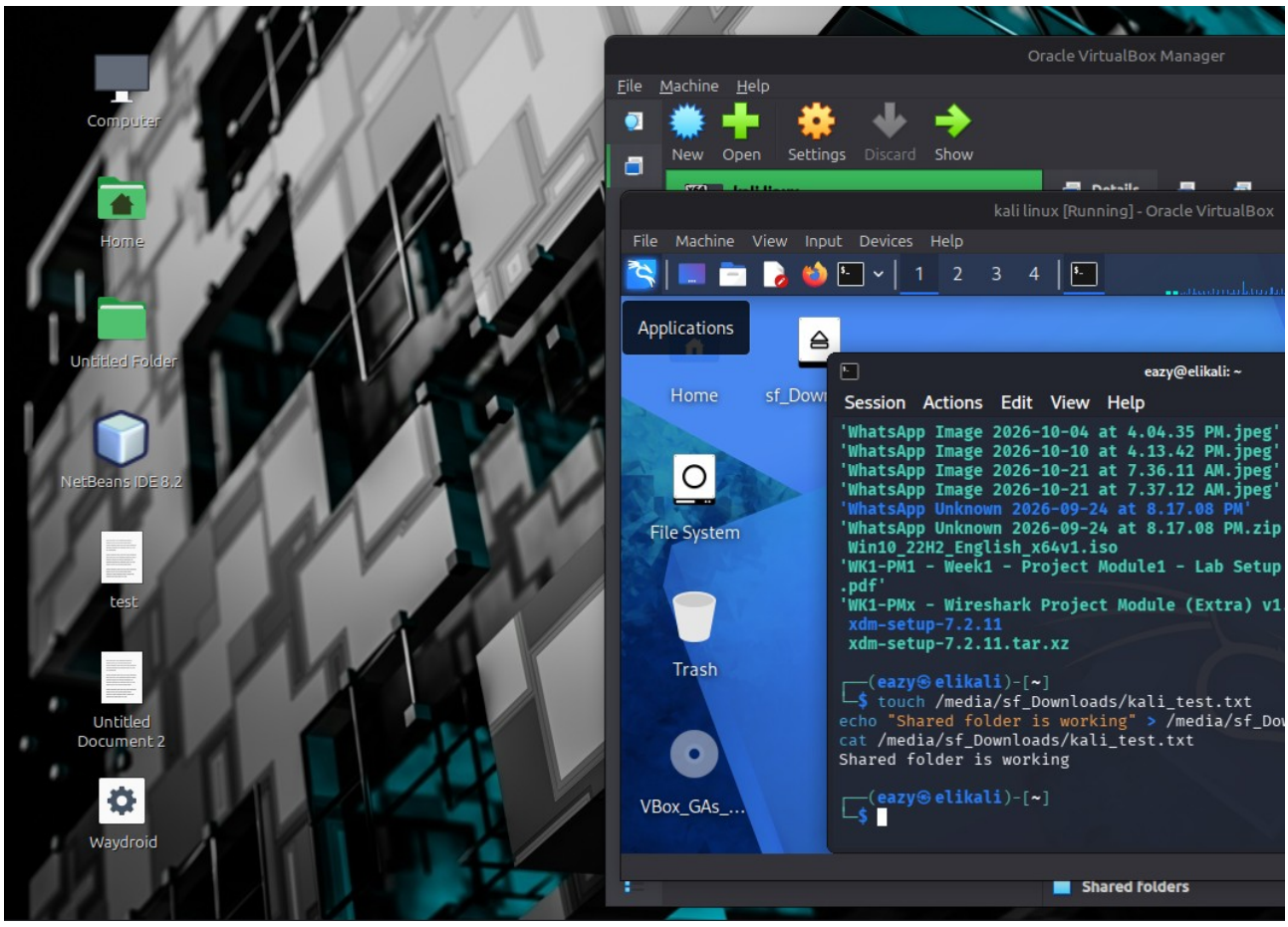
Text was written into the file:

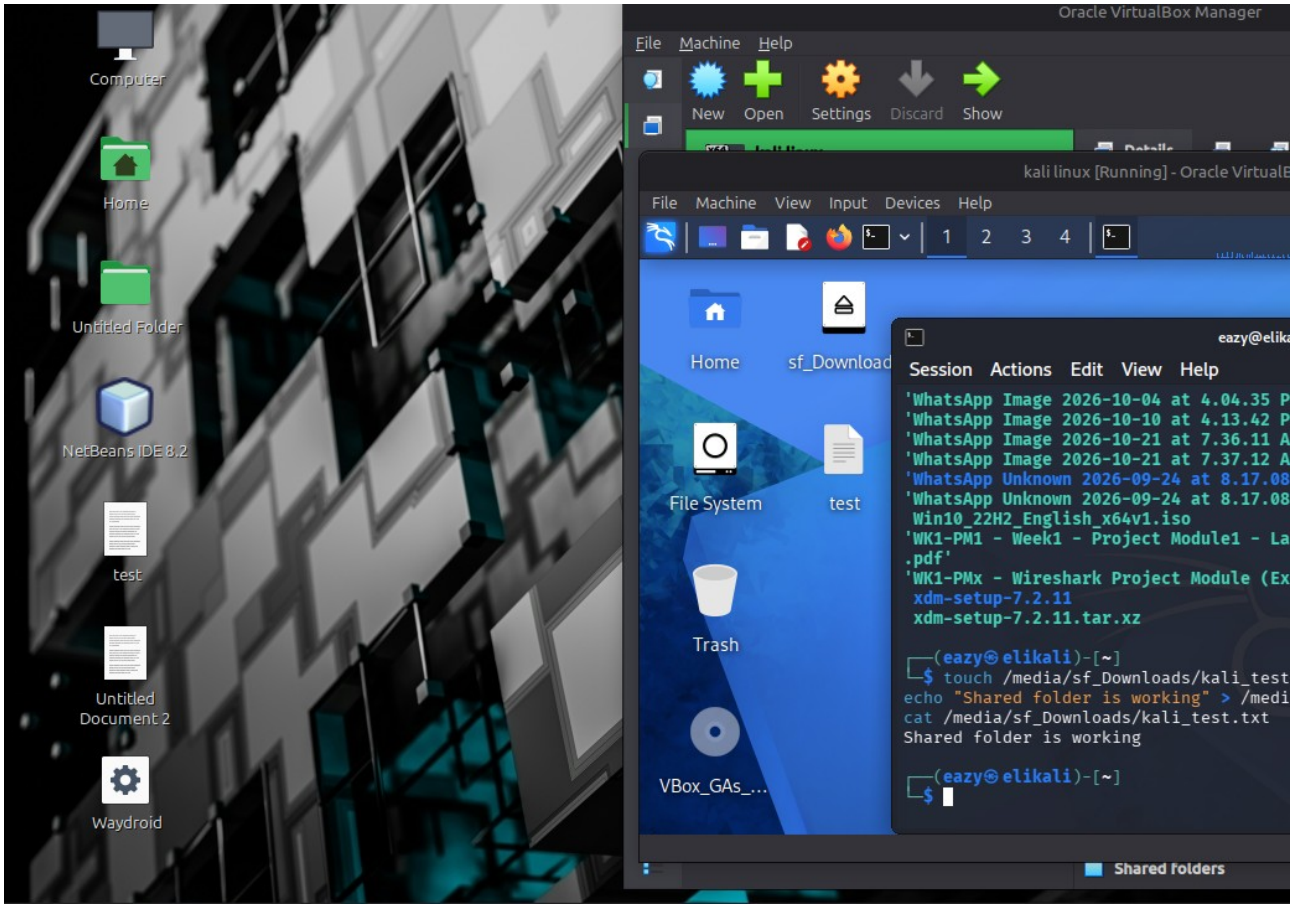
```
echo "Shared folder is working" > /media/sf_Downloads/kali_test.txt
```

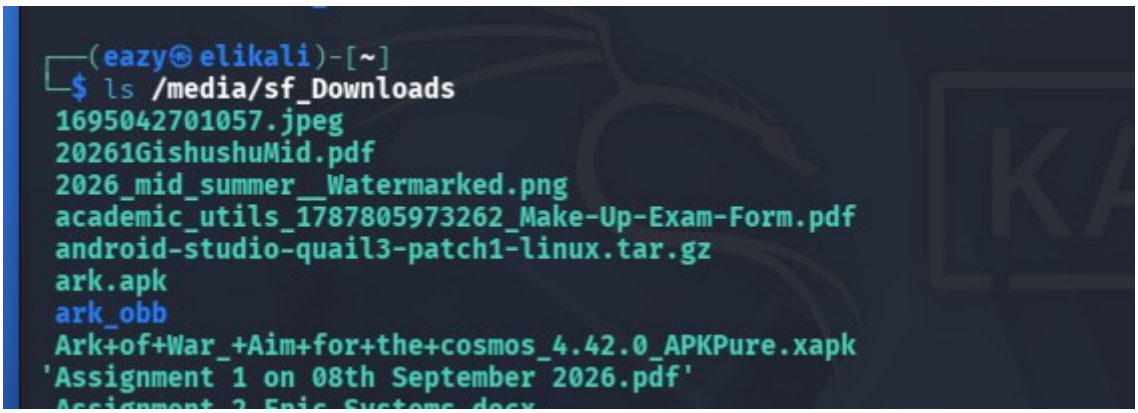
The file was then read using:

```
cat /media/sf_Downloads/kali_test.txt
```

The shared folder was successfully accessed.


Computer
Home
Untitled Folder
NetBeans IDE 8.2
test
Untitled Document 2
Waydroid
Oracle VirtualBox Manager
File Machine Help
New Open Settings Discard Show
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Applications
Home
File System
Trash
VBox_GAs_...
eazy@elikali: ~
Session Actions Edit View Help
'WhatsApp Image 2026-10-04 at 4.04.35 PM.jpeg'
'WhatsApp Image 2026-10-10 at 4.13.42 PM.jpeg'
'WhatsApp Image 2026-10-21 at 7.36.11 AM.jpeg'
'WhatsApp Image 2026-10-21 at 7.37.12 AM.jpeg'
'WhatsApp Unknown 2026-09-24 at 8.17.08 PM'
'WhatsApp Unknown 2026-09-24 at 8.17.08 PM.zip
Win10_22H2_English_x64v1.iso
'WK1-PM1 - Week1 - Project Module1 - Lab Setup
.pdf'
'WK1-PMx - Wireshark Project Module (Extra) v1
xdm-setup-7.2.11
xdm-setup-7.2.11.tar.xz
(eazy㉿elikali)-[~]
$ touch /media/sf_Downloads/kali_test.txt
echo "Shared folder is working" > /media/sf_Dou
cat /media/sf_Downloads/kali_test.txt
Shared folder is working
(eazy㉿elikali)-[~]
$
Shared folders


Computer
Home
Untitled Folder
NetBeans IDE 8.2
test
Untitled Document 2
Waydroid
Oracle VirtualBox Manager
File Machine Help
New Open Settings Discard Show
kali linux [Running] - Oracle VirtualB
File Machine View Input Devices Help
1 2 3 4
Home
sf_Download
File System
test
Trash
VBox_GAs_...
eazy@elika
Session Actions Edit View Help
'WhatsApp Image 2026-10-04 at 4.04.35 P
'WhatsApp Image 2026-10-10 at 4.13.42 P
'WhatsApp Image 2026-10-21 at 7.36.11 A
'WhatsApp Image 2026-10-21 at 7.37.12 A
'WhatsApp Unknown 2026-09-24 at 8.17.08
'WhatsApp Unknown 2026-09-24 at 8.17.08
Win10_22H2_English_x64v1.iso
'WK1-PM1 - Week1 - Project Module1 - La
.pdf'
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xdm-setup-7.2.11
xdm-setup-7.2.11.tar.xz
(eazy㉿elikali)-[~]
$ touch /media/sf_Downloads/kali_test
echo "Shared folder is working" > /medi
cat /media/sf_Downloads/kali_test.txt
Shared folder is working
(eazy㉿elikali)-[~]
$
Shared folders



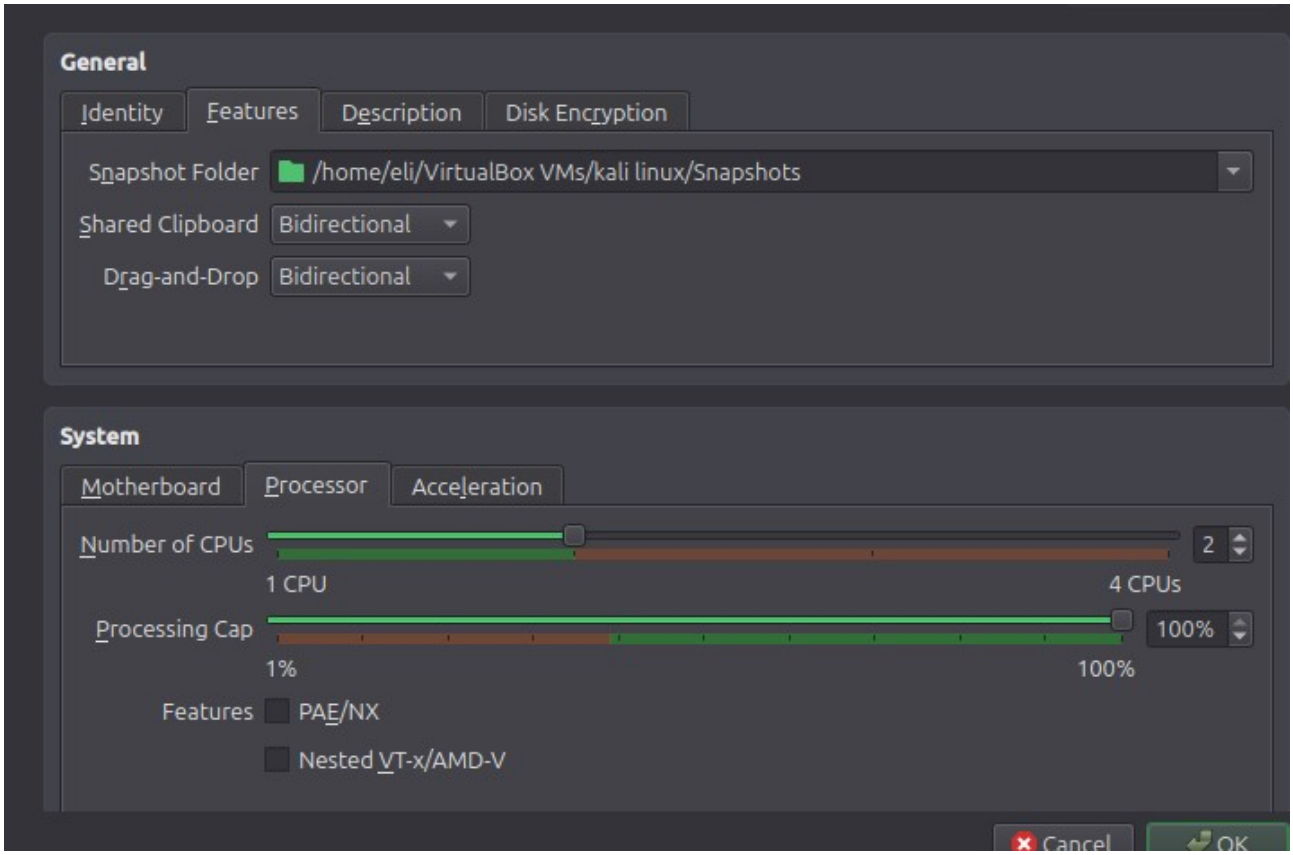
4.7 Clipboard and drag-and-drop configuration

VirtualBox was configured to allow communication between the host operating system and Kali Linux through clipboard sharing and drag-and-drop.

The following settings were enabled:

- Shared Clipboard: Bidirectional.
- Drag and Drop: Bidirectional.

These features make it easier to transfer text and files between the host and the virtual machine.



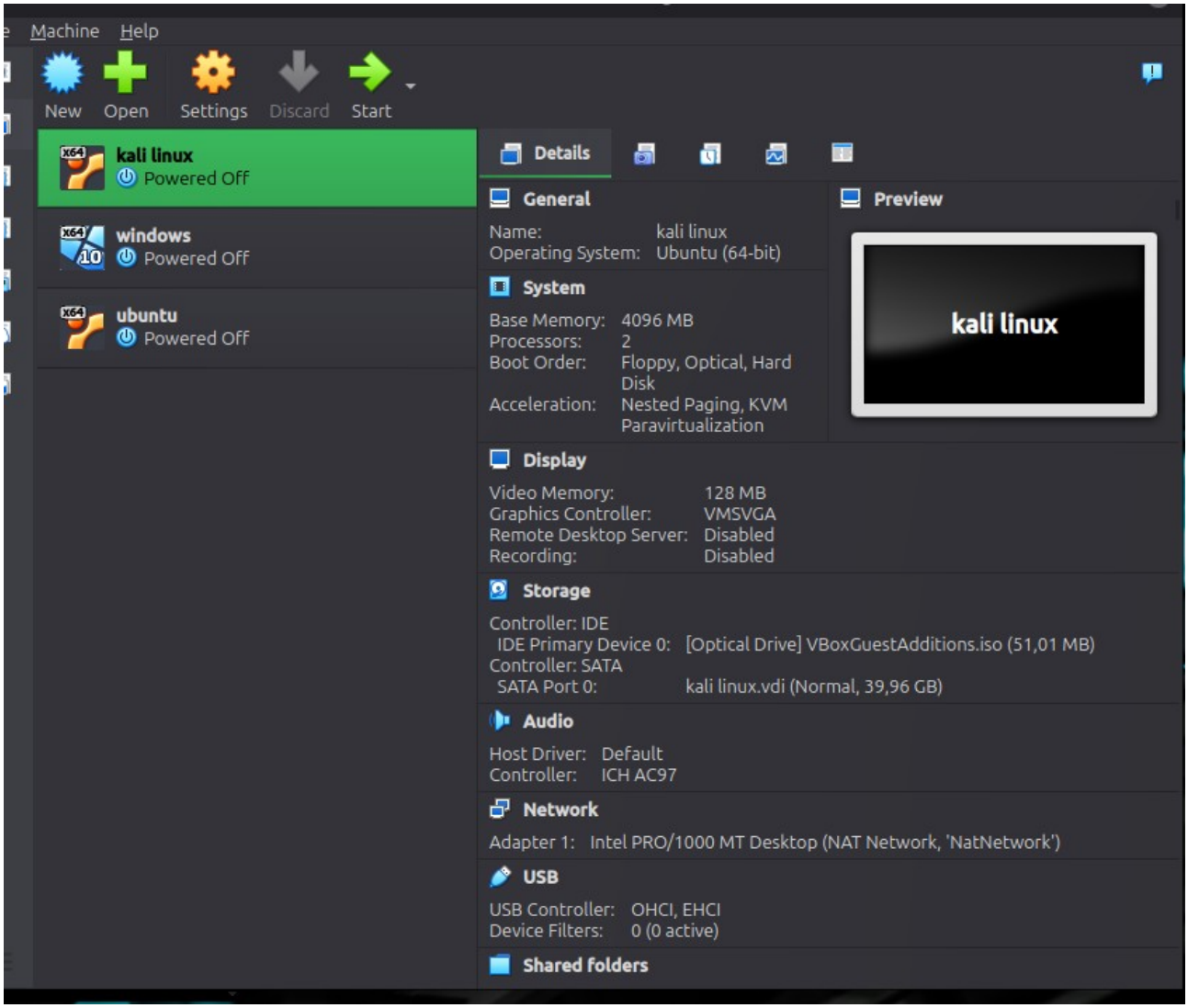
4.8 Creating a Kali Linux snapshot

A snapshot was created to preserve the configured Kali Linux environment.

The snapshot provides a restore point that can be used to return the virtual machine to its saved configuration if future experiments cause unwanted changes.

The snapshot was named:

KaliLinuInitConfi



5. Phase Two: Windows 10 Virtual Machine Setup

The second phase involved installing Windows 10 as an additional virtual machine. This was an optional extension of the original assignment.

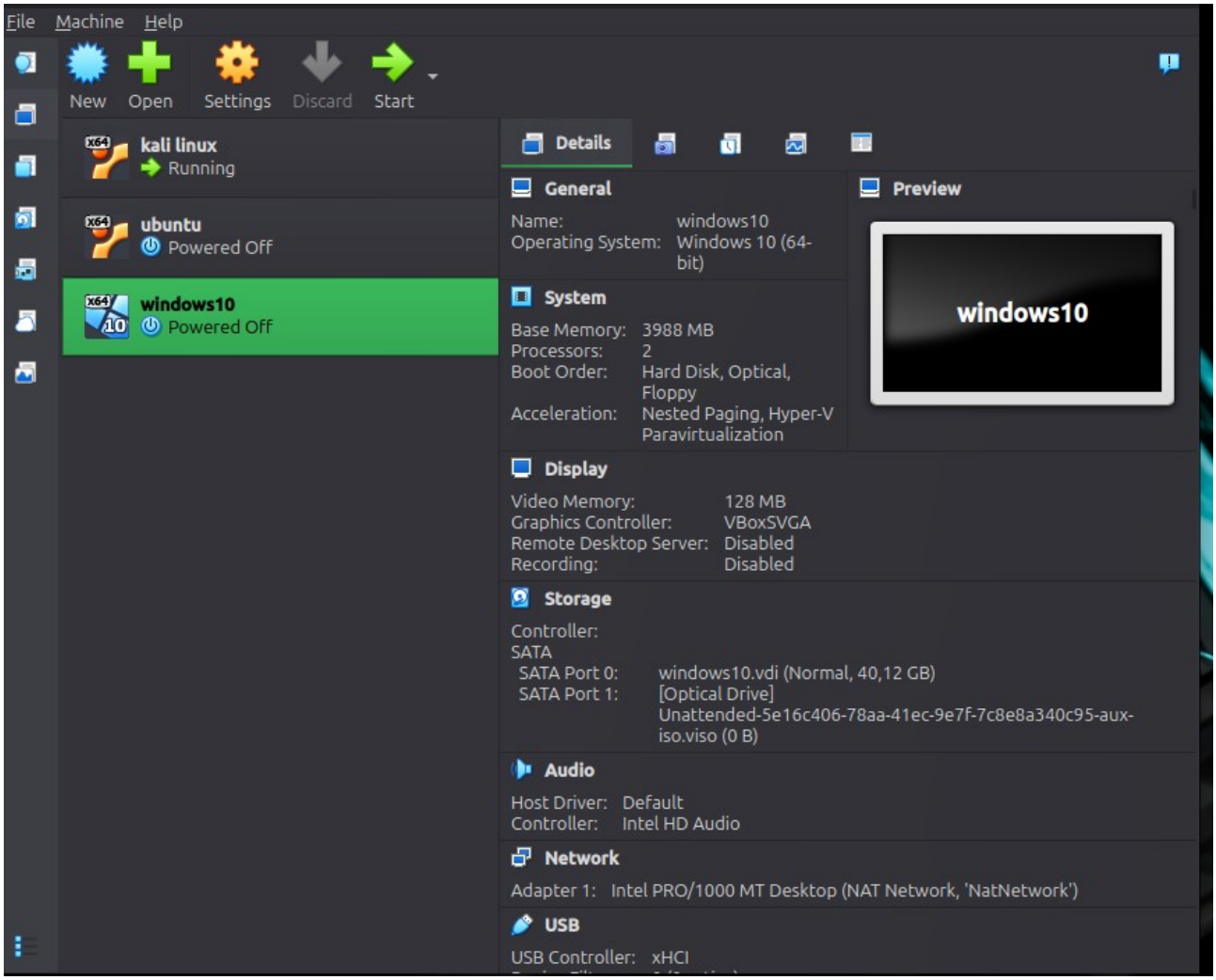
Windows 10 was selected to act as a target machine for testing communication with Kali Linux.

5.1 Windows 10 installation

A Windows 10 ISO image was used to create a new virtual machine in VirtualBox.

The virtual machine was configured with appropriate hardware resources and a virtual hard disk before Windows 10 was installed.

After installation, Windows 10 was started successfully.



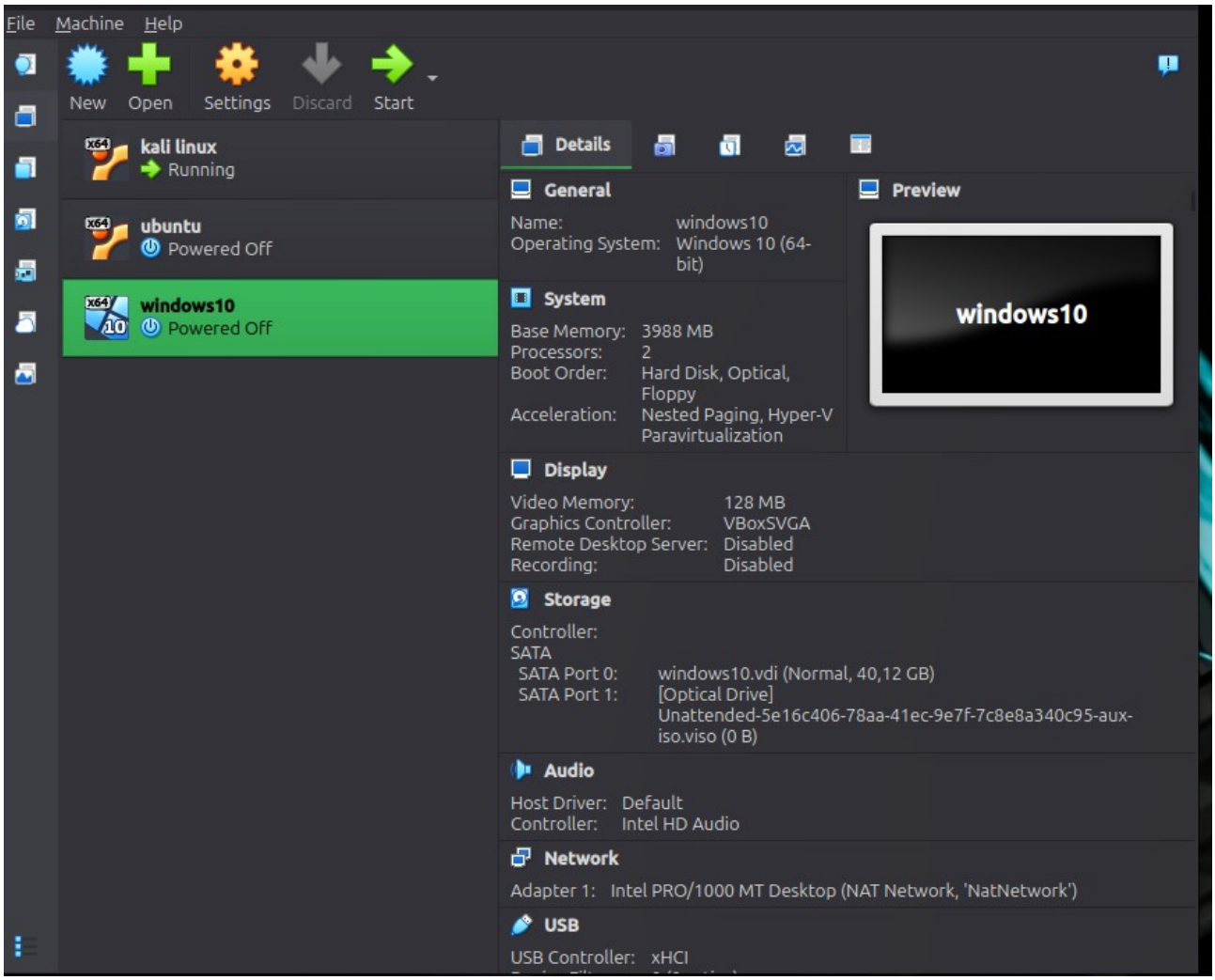
5.2 Configuring the Windows 10 network adapter

The Windows 10 virtual machine was connected to the same custom NAT Network used by Kali Linux.

The adapter was configured as follows:

- Enable Network Adapter: Checked.
- Attached to: NAT Network.
- Network name: NatNetwork.
- Cable Connected: Checked.

This allowed Windows 10 to participate in the same virtual network as Kali Linux.



5.3 Assigning a static IP address to Windows 10

Windows 10 was assigned the static IP address 10.0.0.10/24.

The configuration was performed through the Windows network adapter settings.

The steps followed were:

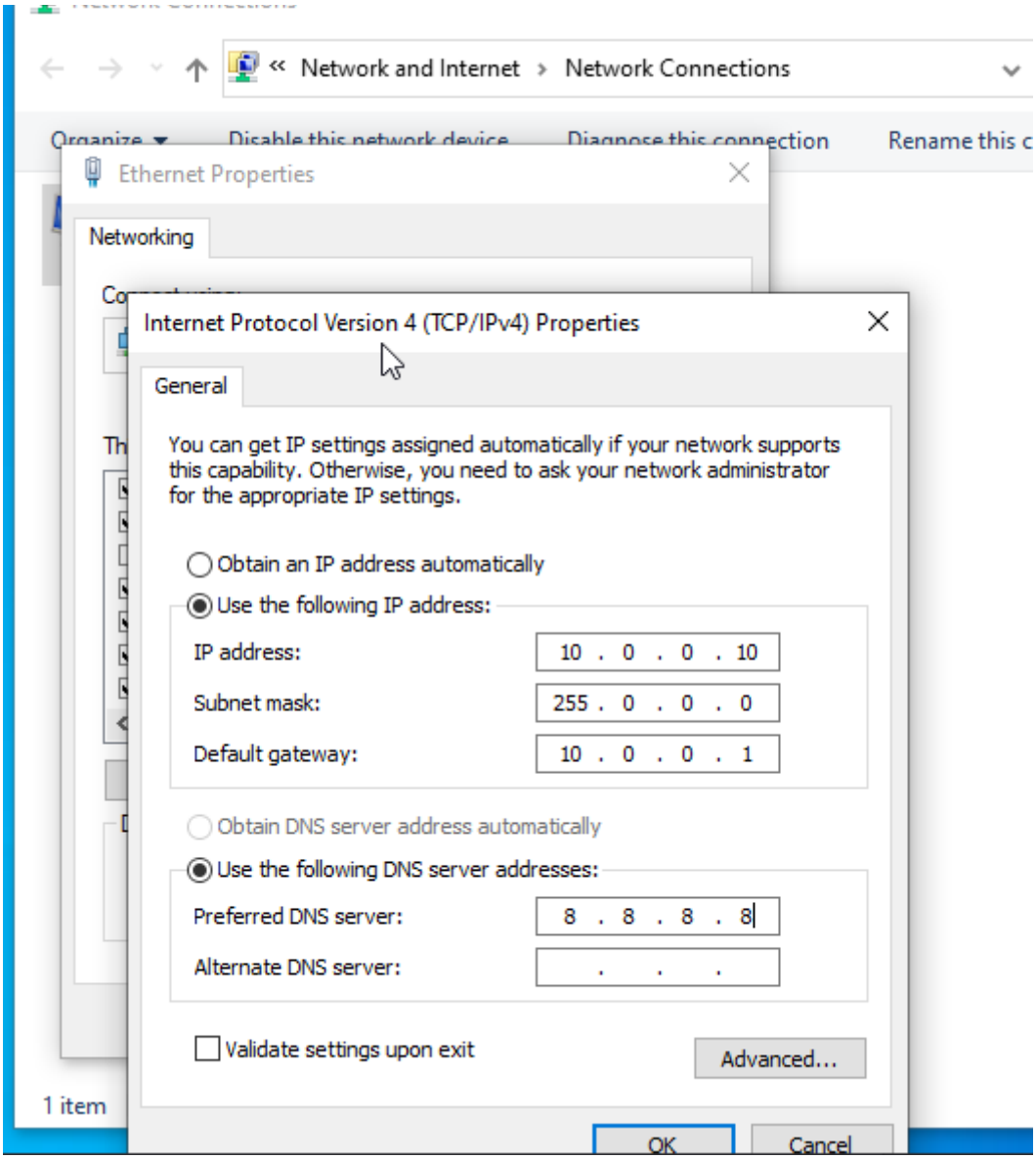
1. Open the Run dialog using Windows + R.
2. Enter ncpa.cpl.
3. Open the Ethernet adapter properties.
4. Select Internet Protocol Version 4 (TCP/IPv4).
5. Select the option to use a static IP address.
6. Enter the network configuration.

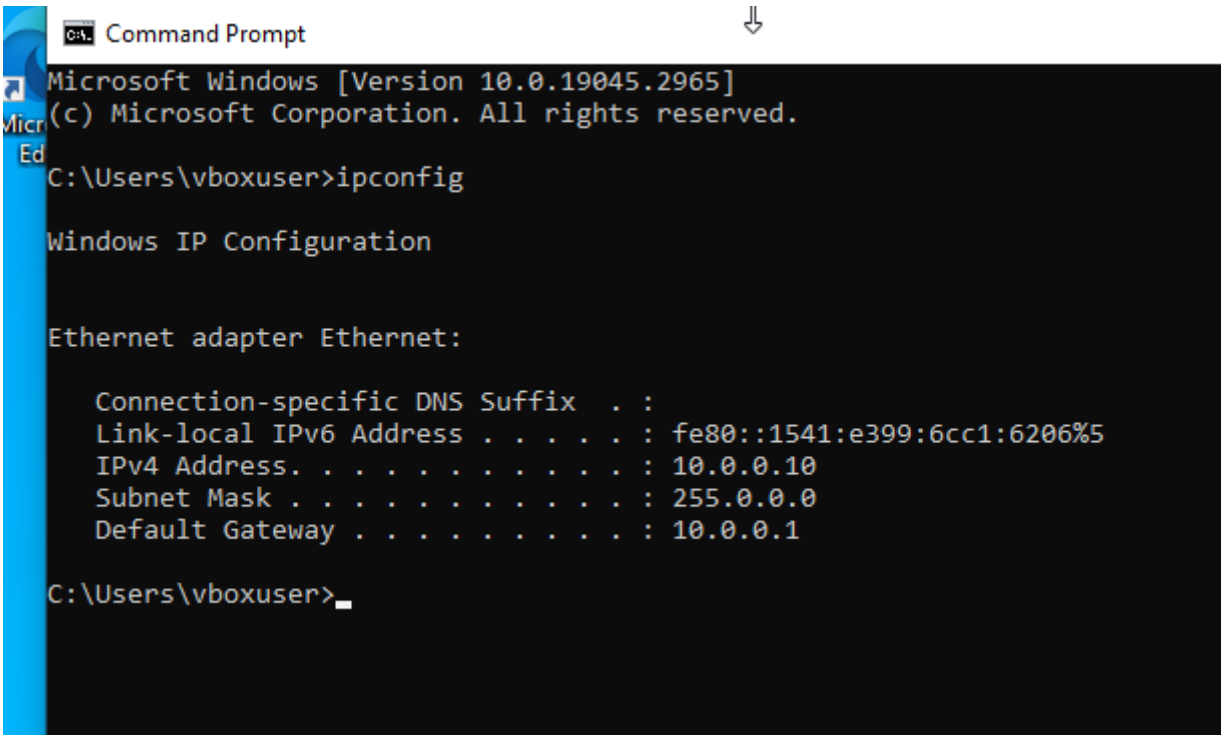
The settings were:

Setting	Value
Method	Manual
IP address	10.0.0.2
Netmask	255.0.0.0
Gateway	10.0.0.1
DNS server	8.8.8.8

The configuration was checked using the following command in Command Prompt:

ipconfig





5.4 Testing communication between Kali Linux and Windows 10

After configuring both virtual machines, network communication was tested using the ping command.

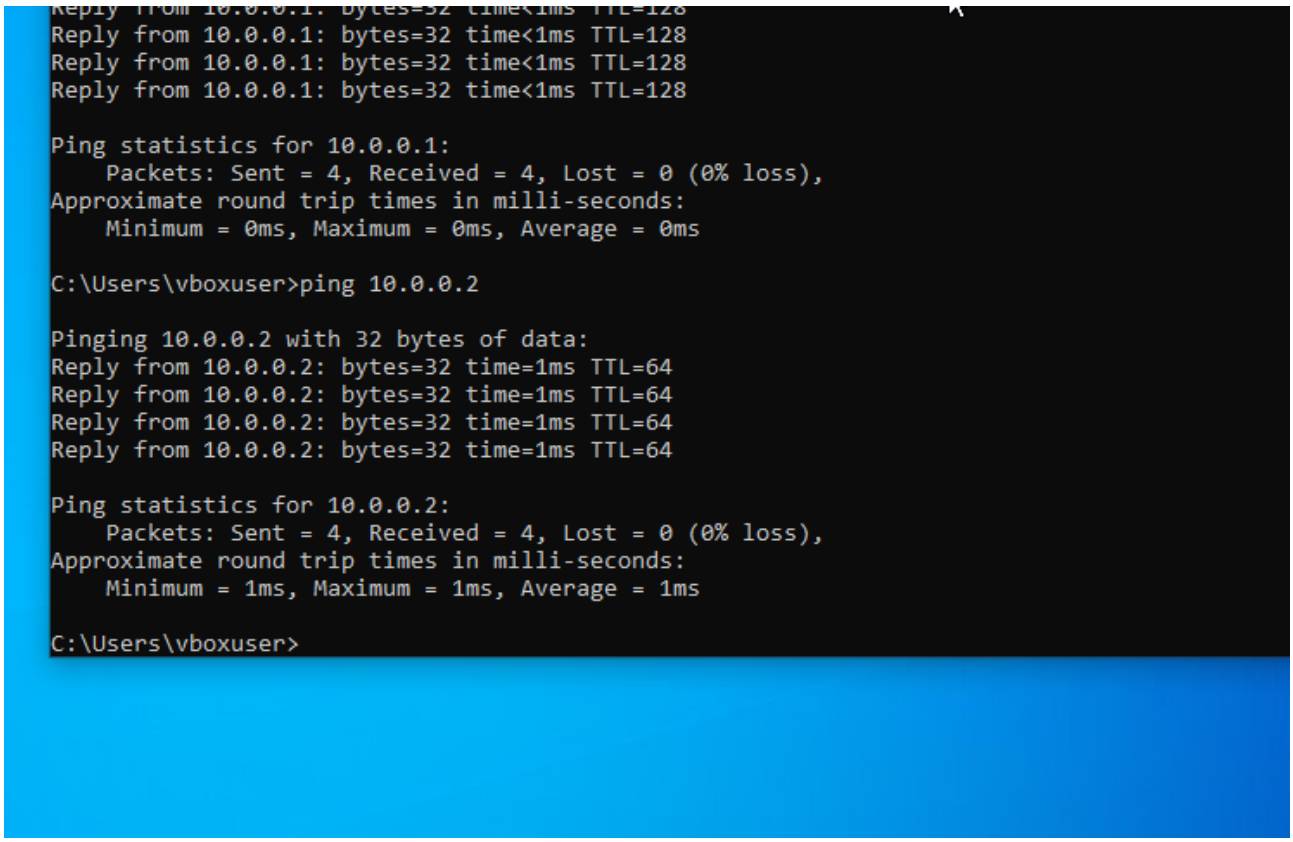
First, the Windows 10 machine was tested from Kali Linux:

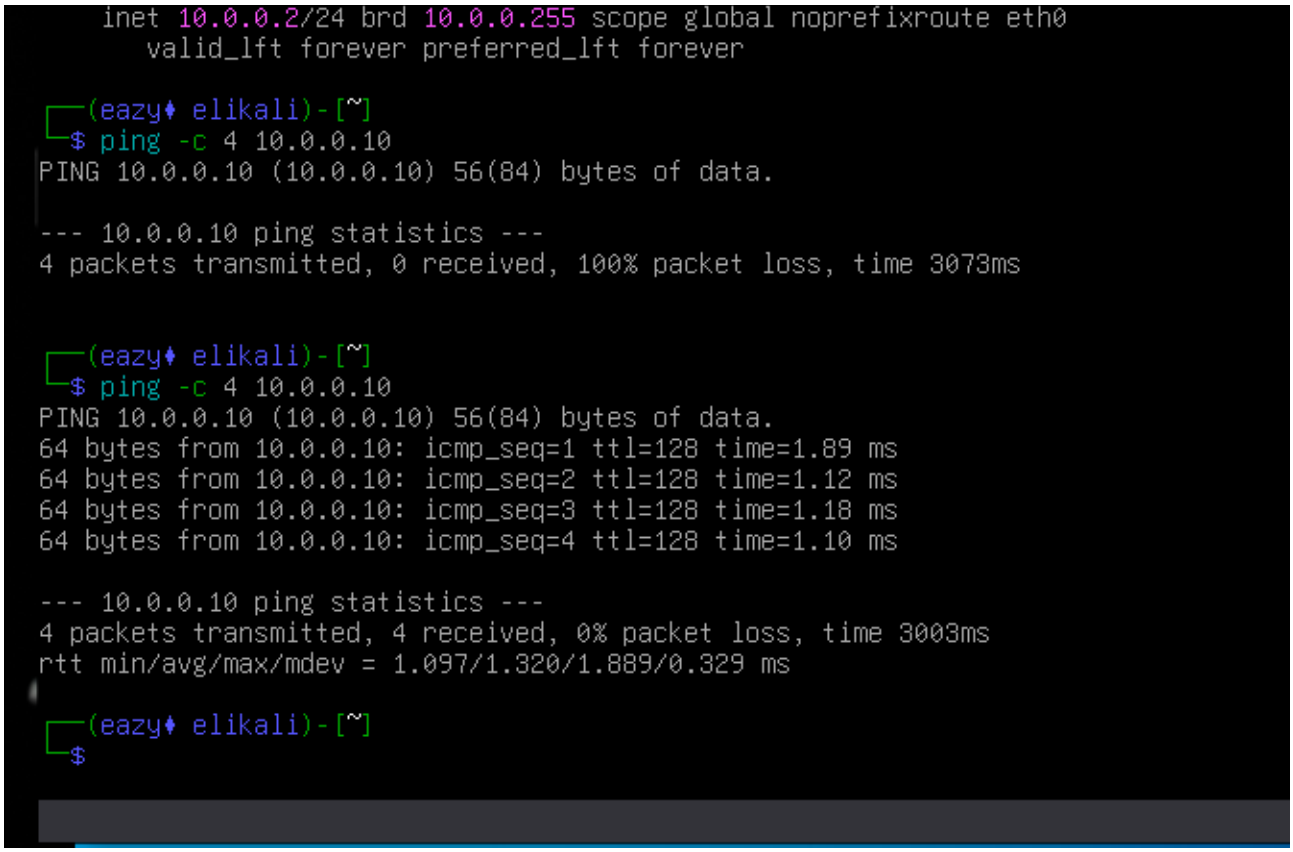
```
ping -c 4 10.0.0.10
```

Next, Kali Linux was tested from Windows 10 using Command Prompt:

```
ping 10.0.0.2
```

Both tests were successful. This confirmed that the two virtual machines could communicate with each other through the configured NAT Network.


Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\vboxuser>ping 10.0.0.2
Pinging 10.0.0.2 with 32 bytes of data:
Reply from 10.0.0.2: bytes=32 time=1ms TTL=64
Reply from 10.0.0.2: bytes=32 time=1ms TTL=64
Reply from 10.0.0.2: bytes=32 time=1ms TTL=64
Reply from 10.0.0.2: bytes=32 time=1ms TTL=64
Ping statistics for 10.0.0.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\Users\vboxuser>


inet 10.0.0.2/24 brd 10.0.0.255 scope global noprefixroute eth0
valid_lft forever preferred_lft forever
(eazy♦ elikali)-[~]
$ ping -c 4 10.0.0.10
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.
--- 10.0.0.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3073ms
(eazy♦ elikali)-[~]
$ ping -c 4 10.0.0.10
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.
64 bytes from 10.0.0.10: icmp_seq=1 ttl=128 time=1.89 ms
64 bytes from 10.0.0.10: icmp_seq=2 ttl=128 time=1.12 ms
64 bytes from 10.0.0.10: icmp_seq=3 ttl=128 time=1.18 ms
64 bytes from 10.0.0.10: icmp_seq=4 ttl=128 time=1.10 ms
--- 10.0.0.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.097/1.320/1.889/0.329 ms
(eazy♦ elikali)-[~]
$

6. Final Lab Configuration

At the end of the practical work, the lab consisted of two virtual machines connected through the same NAT Network.

The final configuration was:

Component	Final configuration
Host operating system	Linux Mint
Virtualization platform	Oracle VirtualBox
Attacking machine	Kali Linux
Kali IP address	10.0.0.2/24
Target machine	Windows 10
Windows 10 IP address	10.0.0.10/24
Network type	NAT Network
Network name	NatNetwork
Network subnet	10.0.0.0/24
Default gateway	10.0.0.1
Shared folder	/media/sf_Downloads
Internet access	Working on Kali
Inter-machine communication	Successfully tested

7. Challenges Encountered and Solutions

During the configuration process, a few issues were encountered.

1. Kali Linux internet connectivity

Initially, Kali Linux did not have internet access. The network configuration was adjusted, and connectivity was restored.

2. VirtualBox Guest Additions and shared folder access

The shared folder was initially difficult to access because the expected mount location was not immediately visible. After configuring the shared folder and its guest support, it became accessible through /media/sf_Downloads.

3. Finding the shared folder location

An incorrect directory was initially entered in the terminal:

```
ls /madia/
```

The correct command was:

```
ls /media/
```

The correct directory was then located.

4. Configuring communication between the virtual machines

Both virtual machines needed to use the same NAT Network and subnet. After assigning the correct static IP addresses, successful ping tests confirmed communication in both directions.